

for IoT based applications like smart grids, smart energy meters. and solar pumps is humongous.

Adding to this, autonomous vehicles are also a new and exciting sector that is slowly taking shape. Even while the world parades over electric vehicles, it is clear that connected mobility is emerging as a star of the show. As the frenzy for connected mobility slowly picks up pace, automotive companies would be on the lookout for smart component makers in the IoT field to help them achieve their vision. With India being a leading player in the automotive industry, making use of the 'first mover advantage' would ensure both growth and sustainability to all parties involved.

At the same time, industrial IoT is also booming considerably as most industries have now identified how a connected workspace can assist the growth of their business. With the help of IoT, many companies are skyrocketing their method of work. The good news is that every industry has a unique requirement, and no company will have a standard module to get into the industrial IoT space. This would force them to look for SMEs and startups to customise from the given requirements.

Challenges

Even while the IoT industry is piled up with opportunities, only a few startups end up making the cut and being able to scale beyond expectations. Keskar breaks down some of the generic issues startups often face.

The first and the foremost is that startups lack a good understanding of all the technology options available for their product, which eventually leads them to doom when it goes out into the market. "After working for a year or two, they realise that their product is neither meeting the commercial expectations of the end market nor it is meeting the technical requirements. Unfortunately, by then it is too late," he adds.

As we know, an IoT product has



a set of basic building blocks necessary for its functioning—the device, connectivity module, and the cloud. The crucial components for the product are semiconductors, printed circuit boards (PCBs), and surface mount devices (SMDs).

Keskar notes, "If you ask me, the most important aspect in manufacturing is that we need to focus on creating a high value-added ecosystem."

He goes on to say, "If you just do electronic manufacturing service (EMS) activity, only 10-15% value of that electronic product is realised. However, analysing the value chain of any electronic product shows that IP and technology contribute 40-50% and components like PCBs, chips, displays, etc contribute only 30% value to the value chain." Unfortunately, in terms of semiconductors, our country is far from being self-reliant given how dependent we are on imports.

"It should be worrying us because as an industry I believe everybody has accepted this status quo, that we are going to be dependent on imports coming from China, Korea, or Taiwan for even small components like SMD resistors. So, the major question here is that, can we really create this \$20 billion opportunity while being dependent on imports?" asks Arpit Chhabra, CEO & Co-founder, IoTfy.

Even while the government has recently shown interest in building the country to be part of the chip manufacturing crowd, we can assume that it would take a substantial amount of time to completely

quit relying on imports and truly become self-reliant first even before thinking of exports.

Chhabra adds that the missed opportunity, when the chip industry across the globe was still nascent, set us back in years. He says, "At this stage, we are stuck because for the last 15 years we couldn't move. Companies like China, Korea, and Taiwan have really scaled up their capacities during that time and now supply to countries globally."

Solution

India, today, either acts as a design centre or service unit for global companies, but there are very few product companies. This implies the need of the hour is creating successful product innovation companies from India. "It is in the national interest that unless we create for the local market and global market, we are not going to recognise that 40-50% value, which product technology or IP amounts to. So, our focus has to be on creating local product innovation," emphasises Keskar.

To achieve this, a strong policy framework that supports the vision and goes hand-in-hand with the demands and requirements of the industry is needed. "A semiconductor fab is an expensive affair, which is why I believe that the government must look at it from a 10-15 year timeframe, so that we know that all the investments that we are doing on the policy side are really giving us the returns that we're looking at," suggests Chhabra.

Additionally, providing single-window clearances for investments